Higher National Diploma in Engineering (Electrical)

First Year, Second Semester Examination - 2022

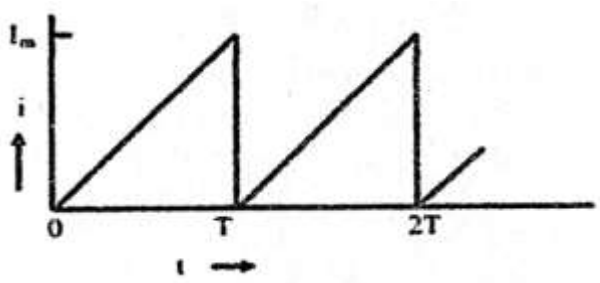
22EN1212 Electrical and Electronic Engineering Fundamentas II

Answer any 5 questions Time : 3 hours

Question 1

(i) Write the mathematical expression for the average and root mean square (r.m.s.) calculation for a random a.c. signal. (4 marks)

(ii) Obtain the average and r.m.s. values for the triangular current waveform are given below. (6 marks)



Triangular current waveform

(iii) An alternating voltage is given by

$\mathbf{V(t) = 325\sin(314.28t)}$ Volt. Find: (a) **Peak to Peak value** (b) **Root Mean Square (r.m.s.) voltage** (c) **Frequency** (d) **Instantaneous value of voltage** when $\mathbf{t = 3\ ms}$ (4 x 2 = 8 marks)

(iv) Two a.c. voltages are given by $\mathbf{V_1(t) = 120\sin(\omega t)}$ and $\mathbf{V_2(t) = 200\sin(\omega t - \pi/4)}$ volts. Obtain the **sinusoidal expression for $\mathbf{V_1(t) + V_2(t)}$ by resolution of phasors**. Whether $\mathbf{V_2(t)}$ is **leading or lagging $\mathbf{V_1(t)}$**? Give the reason for your answer. (4 marks) (3 marks)

Question 2

(i) A $40\ \mu F$ **capacitor** is connected in **parallel with a** $100\ \Omega$ **resistor** across a $\mathbf{240\ V, 50\ Hz}$ supply. Calculate the following: (a) The **current in each branch** (b) The **supply current** (c) The **circuit phase angle** between supply voltage and current (d) The **circuit impedance** (e) The **power dissipated** in the circuit (f) The **apparent power** in the circuit (g) The **power factor** of the supply (14 marks) **(ii)** Define the following for resonant circuits: (a) **Resonance frequency (** $\mathbf{f_r}$**)** (b) **Quality factor (**$\mathbf{Q}$**-factor)** (c) **Bandwidth (**$\mathbf{BW}$**)** (3 marks) **(iii)** A coil having a **resistance of** $\mathbf{20\ \Omega}$ and an **inductance of** $\mathbf{200\ mH}$ is connected in **series with a** $\mathbf{60\ \mu F}$ **capacitor** across a $\mathbf{230\ V}$ supply. (a) At what **frequency** does **resonance** occur? (b) Find the **current flowing at the resonance frequency**. (c) Determine the **bandwidth** and $\mathbf{Q}$**-factor** of the circuit. (8 marks)

Question 3

(i) (a) What are the **advantages of a three-phase system over a single-phase system**? (State two advantages.) (b) **Compare the Star and Delta connections** in the 3-phase system. (4 marks) **(ii)** Three coils have a **resistance of** $\mathbf{5\ \Omega}$, and an **inductive reactance of** $\mathbf{10\ \Omega}$ and are connected in **star** to a $\mathbf{400\ V}$, **3-phase supply**. Calculate the following: (a) The **line and phase voltages** (b) The **line and phase currents** (c) **Apparent power** delivered by the supply (d) **Power factor** of the load (e) How do you measure the **active power** in a 3-phase system, using the **two-wattmeter method**? (f) What are the **readings of each wattmeter** for the above 3-phase circuit? (11 marks) **(iii)** (a) State two **advantages** and one **disadvantage** of the **PMMC (Permanent Magnet Moving Coil) instrument** used in the construction of voltmeters and ammeters. (b) Briefly explain the **accuracy** and **precision** of the measurement of electrical quantities. (6 marks) **(iv)** A **moving-coil instrument** gives **full-scale deflection with** $\mathbf{20\ mA}$ and has a **resistance of** $\mathbf{10\ \Omega}$. Calculate the **resistance required**: (a) in **parallel** to enable the instrument to read up to $\mathbf{1\ A}$. (b) in **series** to enable it to read up to $\mathbf{10\ V}$. (4 marks)

Question 4

(i) (a) Briefly explain the **necessity of a transformer** in the power system. (b) How does a **transformer operate**? (c) What is the **advantage of using an isolation transformer** in the power circuits? (d) What are the **losses that took place in the transformer** while operating on load? Briefly explain them. (e) Define the **voltage regulation of the transformer**. Why is it important in the transformer? (8 marks) **(ii)** (a) What are the **common losses in any type of rotating machine**? (b) What is the **condition to be satisfied by any type of rotating machine to operate at its maximum efficiency**? (c) State the **difference between a D.C. generator and a D.C. motor**. (d) **Compare the A.C. synchronous machine with the Induction machine**. (6 marks) **(iii)** A $5\ kVA$ **single-phase transformer** has a **turn ratio of** $10:1$ and is fed from a $2\ \mathbf{kV}$ supply. Calculate the followings: (a) **Approximate values of primary and secondary currents**. (b) The **m.m.f (Magnetic motive force) of the secondary** which has $80\ \mathbf{turns}$ of winding. (c) The **approximate number of turns in the primary winding**. (d) The **minimum load resistance** that can be connected across the secondary winding to get full load kVA. (e)

What is the **voltage regulation of the transformer** if the voltage at load is $195\ \mathbf{V}$? (7 marks) **(iv)** (a) A **6-pole d.c. generator** has a **lap-wound armature with** $50\ \mathbf{slots}$ with $16\ \mathbf{conductors}$ **per slot**. The **useful flux per pole is** $25\ \mathbf{mWb}$. Determine the **speed** at which the machine must be driven to generate an **e.m.f. of** $220\ \mathbf{V}$. (b) What is the **speed of an a.c. machine in revolution per minute (rpm)** with $6\ \mathbf{poles}$ and operate at $50\ \mathbf{Hz}$? (4 marks)

Question 5

(i) (a) Convert the following **binary numbers** into their **hexadecimal and decimal equivalent**: (I) $\mathbf{10101011101}$ (II) $\mathbf{1001010111}$ (b) **Decode the following BCD numbers**: (I) $\mathbf{01000001\ 01110000\ 00010010\ 10010001\ 10000110}$ (II) $\mathbf{01100001\ 01010001\ 01010000\ 01010000\ 00000001}$ (c) **Distinguish between a binary and a BCD code**. Why BCD codes are used for decimal numbers in digital systems? **(ii)** (a) Using the **theorems of Boolean algebra**, prove the following **identities**: (I) $\mathbf{A}\cdot\overline{\mathbf{B}}+\overline{\mathbf{A}}\cdot\mathbf{B}=\overline{\mathbf{A}\cdot\mathbf{B}+\overline{\mathbf{A}}\cdot\overline{\mathbf{B}}}$ (II) $\overline{\mathbf{X}}\cdot(\mathbf{X}+\mathbf{Z})+\overline{\mathbf{Y}}+\mathbf{Y}\cdot\mathbf{Z}=\overline{\mathbf{Y}}+\mathbf{Z}$ (III) $\mathbf{A}\cdot\mathbf{B}+\mathbf{A}\cdot\mathbf{C}+\overline{\mathbf{B}}\cdot\mathbf{C}=\mathbf{A}\cdot\mathbf{B}+\overline{\mathbf{B}}\cdot\mathbf{C}$ (6 marks) (b) Using the **theorems of Boolean algebra, reduce the following function** and **draw the logic circuit** using **AND, OR and NOT gates only**:

$$\mathbf{F_1(A, B, C, D) = \Sigma(0, 1, 2, 6, 7, 14, 15)}$$

(6 marks) **(iii)** (a) What are the **differences between conventional and non-conventional sources of energy**? (b) Briefly explain the **electric shock** and the **causes of electric shock**. (c) Briefly explain the **earthing systems** used in electrical wiring. (5 marks) **(iv)** (a) What is **IEE wiring regulations** and the **latest version of these regulations**? (b) What are the **basic steps or requirements in wiring a residential house**? (c) What is the **difference between RCCB and MCB**? (d) Why do we need a **2-way switch in-house electrical wiring**? Draw the **circuit diagram for a 2-way switch** in detail. (8 marks)